Focus Panda: An AI Assisted Robotic Companion for Focus and Study Habits

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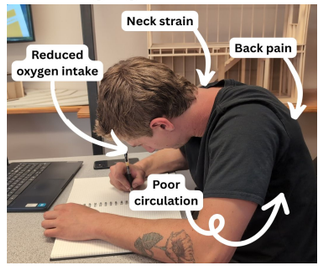
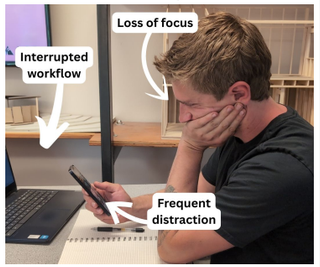
Professor Jacob Wikowsky, MS PER - EGG1040101 Spring 2026

Problem Statement

Students struggle to stay focused during study sessions because of phone distractions, poor posture, and no immediate reward, leading to low engagement and inconsistent study habits.



75% of students have "very poor" posture [1]



70% of students identify as procrastinators!! [2]

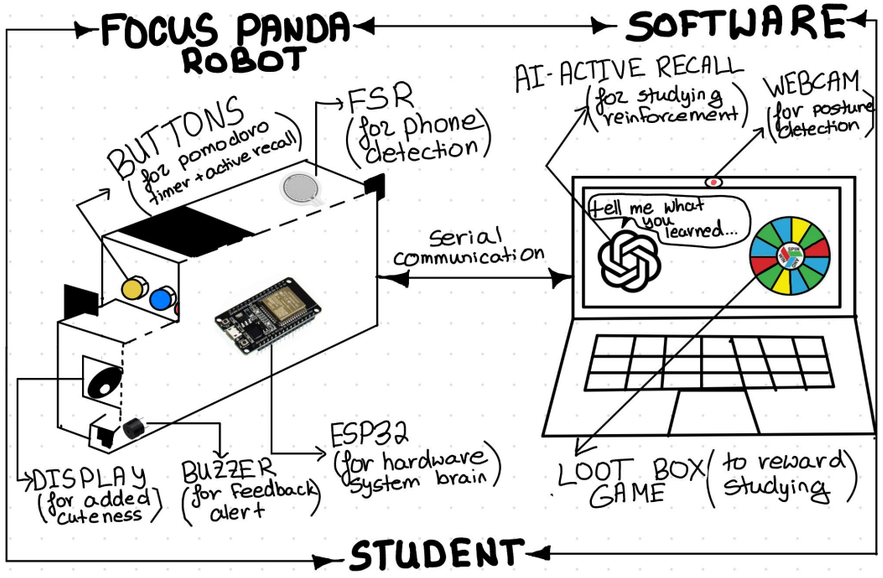
Existing Solution Comparison

Solution	Focus	Posture	Physical Interaction	Reward System
Pomodoro Apps	✅	❌	❌	❌
Posture Systems	❌	✅	❌	❌
AI Robots	❌	❌	✅	❌
Focus Panda	✅	✅	✅	✅

Proposed Solution

Interactive study system designed to reduce distractions, improve posture, and encourage focused studying through phone detection, active recall, and randomized rewards.

System Overview



How It Works

Detect
Phone + posture

↓

Analyze
AI processes posture + active recall

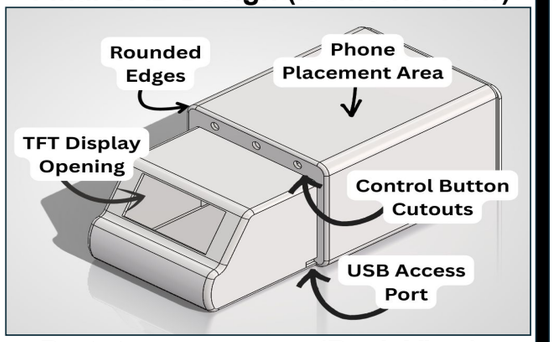
↓

React
Robot gives feedback

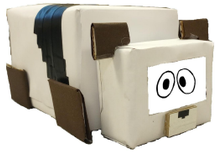
↓

Reward
Coins + variable rewards

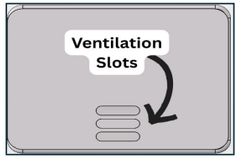
Final CAD Design (Isometric View)



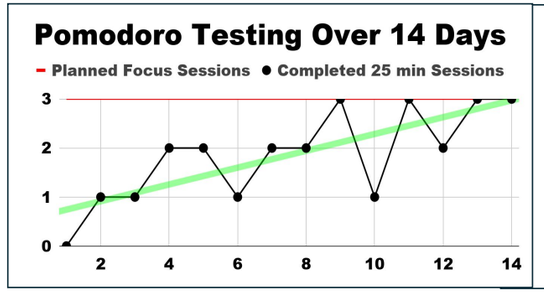
Prototype

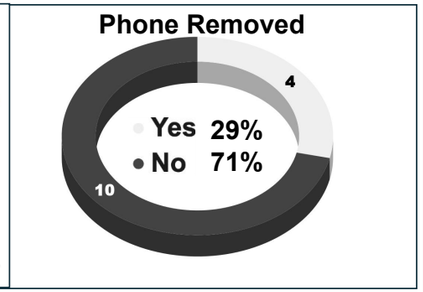


(Back View)



Stakeholder Testing Results





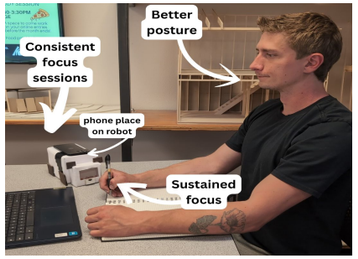
System Validation

- ESP32 ↔ website communication successful
- Reward signals transferred correctly
- Stable timer state transitions
- Randomized rewards generated successfully
- Reliable system operation during testing

Future Improvements

- AI posture detection
- Active recall integration
- Full wireless operation
- Custom 3D printed housing
- Expanded reward system

Benefits



Focus Panda encourages structured studying through physical interaction and reward based motivation.

Costs

Items	Price ($)
ESP32 Microcontroller	10.00
LCD Display (ILI9341)	12.00
Push Buttons (x3)	1.12
Bread Boards + Jumper Wires	5.60
Buzzer Module	1.40
Force Sensor (FSR)	10.00
USB Cable/ Power	3.00
PLA Filament Material	4.50
Small Plastic Hinges	2.00
Cost Per Unit	49.62

Common Questions & Answers

How does the system detect the phone?

The robot uses a force sensitive resistor underneath the phone placement area to detect pressure from the phone.

What happens if the phone is removed?

The system interrupts or resets the study session to discourage distractions.

Why use a physical robot instead of just an app?

Physical interaction and real time feedback make the system more engaging and harder to ignore than digital-only study tools.

How does the reward system work?

The system uses randomized rewards with a 60% chance of winning and 40% chance of losing to help maintain motivation.

How does the website communicate with the robot?

The ESP32 communicates with the website through serial communication.

Is the posture detection fully implemented?

For this prototype, posture detection is simulated to demonstrate the workflow. Full AI posture detection is planned for future development.

Why did you choose ESP32?

The ESP32 is affordable, compact, and supports communication and sensor integration needed for the prototype.

What programming tools did you use?

The project used Arduino IDE for the ESP32, React for the website, and CAD software for the housing design.

What was the biggest challenge?

One of the biggest challenges was integrating the hardware, website, and communication system into one working prototype.

What are future improvements?

Future improvements include AI posture detection, expanded active recall, wireless operation, and improved housing design.

References

1. S. T. Lauman and S. T. Lauman, "Impacts and Attitudes of Posture on Daily Function, Disability, and Psychological Measures in College Students", Journal of Musculoskeletal Disorders and Treatment, vol. 9, no. 2, pp. 124, Jun. 2023, doi: 10.23937/2572-3243.1510124. [Online]. Available: https://clinmedjournals.org/

2. J. R. Ferrari, J. F. Díaz-Morales, J. O'Callaghan, K. Díaz, and D. Argumedo, "Frequent Behavioral Delay Tendencies by Adults", Journal of Cross-Cultural Psychology, Jan. 2007. [Online]. Available: https://journals.sagepub.com/doi/abs/10.1177/0022022107302314